

PAPER 1 / QUESTION 112

29 April 2025 / TZ3 (file label) / HL Paper 1A / original question 2 / 2225-6719

- 112.** Two balls, P and Q, are projected from the edge of a cliff. P is projected horizontally at speed v and Q is projected vertically upwards at the same speed. The time taken for P to reach the ground is t_p and the speed at impact with the ground is v_p . **Air resistance is negligible.**

What is the time taken to reach the ground and the speed at impact with the ground for Q?

	Time taken	Speed at impact
A.	t_p	v_p
B.	t_p	Greater than v_p
C.	Greater than t_p	v_p
D.	Greater than t_p	Greater than v_p